

ReguMind — Literature Review, Feedback & Evaluation, and Grading Criteria

1) Literature Review

ReguMind is grounded in recent advances in Retrieval-Augmented Generation (RAG), semantic search, and document question-answering systems. Traditional large language model (LLM) chatbots rely only on pre-trained knowledge, which often leads to hallucinated or unverifiable answers. Research shows that combining LLMs with external knowledge retrieval significantly improves factual accuracy and trustworthiness.

RAG-based architectures retrieve the most relevant document chunks using vector similarity search before passing them to the LLM for answer generation. This grounding process ensures that responses are directly based on real documents rather than the model's internal memory. Such approaches are particularly effective for long, complex, and formal documents like legal texts and academic regulations.

Hybrid retrieval techniques that combine Dense Embeddings with lexical search methods such as BM25 outperform single retrieval methods. Dense retrieval captures semantic meaning, while BM25 captures exact keyword matching. Re-ranking retrieved results before generation further improves the relevance and quality of answers.

Tools such as SentenceTransformers for embeddings, FAISS/Chroma for vector indexing, and LangChain for orchestration are widely used in building scalable document intelligence systems. Multi-tenant AI architecture is also an important concept applied in ReguMind to ensure complete isolation of university data within the same system.

By combining semantic embeddings, hybrid retrieval, strict grounding, and citation-based answering, ReguMind follows state-of-the-art practices in regulation intelligence and document QA systems.

2) Feedback & Evaluation (Lecturer's Assessment)

The project demonstrates a strong understanding of modern AI system design by applying Retrieval-Augmented Generation to a real-world academic problem. The team successfully implemented hybrid retrieval, citation grounding, and a structured AI pipeline separating preprocessing, retrieval, and generation.

Strengths observed:

- Clear problem definition with practical relevance.
- Proper implementation of hybrid retrieval and re-ranking.
- Reliable citation mechanism to minimize hallucinations.
- Effective integration between AI backend and web interface.

- Good balance between research concepts and practical implementation.

The project reflects solid technical execution within the proposed timeline and shows a clear mapping between research ideas and system design.

3) Suggested Improvements

Although the system provides a functional prototype, several enhancements can further improve ReguMind:

- Add authentication and role-based access control.
- Improve UI/UX design for better usability.
- Support additional document formats beyond PDFs.
- Deploy on cloud infrastructure for scalability.
- Implement continuous evaluation metrics and logging.
- Fine-tune embeddings or re-ranking models on regulation-specific data.

These improvements would help transform the prototype into a production-ready academic assistant.

4) Final Grading Criteria

Component	Marks Distribution
Documentation (Proposal, Planning, Literature Review)	25%
System Implementation (RAG, Retrieval, Backend, Frontend)	35%
Testing & Evaluation (Accuracy, Citations, Performance)	20%
Presentation & Demo	20%